

When Should an Agent Distrust Its Own Context?

Model-Dependent Outcomes of an Audited Reasoning Pipeline

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Abstract

Agents grounded in heterogeneous sources often encounter conflicting evidence. We study an Audited Reasoning Pipeline (ARP): a forensic analyzer reads the raw sources and emits a sanitized diagnostic, while a clean-room synthesis stage receives the question and diagnostic but not the raw context. On 100 corrupted ConflictQA instances per model, ARP raises exact match from 15 to 30 for GPT-OSS-120B and from 19 to 29 for MiniMax-M2.5. CAT-Source changes from 40 to 49 and from 74 to 56, respectively; their paired model-by-condition interaction is 27 points (95% CI [13, 41], $p < .001$), although two model/configuration pairs do not identify capability as its cause. The three diagnostic metrics are emitted before the clean-room boundary, and all items are corrupted, so conflict detection has no false-positive evaluation. Token-F1 also falls on multi-answer questions. Because the clean four-call control that gives Stage 4 both raw context and the diagnostic was not run, the air-gap’s causal effect remains unidentified.

1 Introduction

Knowledge graphs and retrieved passages can directly contradict one another. Such context can derail generation or reinforce a model’s prior belief (Shi et al., 2023; Pan et al., 2023; Xie et al., 2024). We study whether separating evidence inspection from final generation is useful in this setting.

Our Audited Reasoning Pipeline (ARP) combines a closed-book snapshot, an evidence exploration pass, forensic analysis, and clean-room synthesis. Only the final stage cannot read the raw sources. We compare this four-call pipeline with a two-call RAG baseline on ConflictQA (Zhao et al., 2026). Because call count, intermediate inputs, decomposition, and prompts all change, the comparison evaluates end-to-end procedures rather than the air-gap alone; the clean four-call raw-context con-

trol was not run. Moreover, the diagnostic metrics are produced upstream of Stage 4.

ARP improves exact match for both evaluated models, but source attribution falls for MiniMax-M2.5 and does not reliably improve for GPT-OSS-120B. The CAT-Source interaction differs across these configurations, but model identity does not independently measure capability. We also find lower token-F1 on multi-answer questions and emphasize that every benchmark item is corrupted.

2 Related Work

Tool-augmented methods such as ReAct and Self-Ask interleave reasoning with retrieval (Yao et al., 2023; Press et al., 2023). Under knowledge conflicts (Xu et al., 2024), models can follow misinformation, distracting context, or parametric priors (Pan et al., 2023; Shi et al., 2023; Mallen et al., 2023). ConflictQA instantiates KG–text disagreement and proposes XoT, a two-stage explanation-based thinking framework (Zhao et al., 2026). We use its benchmark but do not directly compare against XoT and make no superiority claim; other remedies include context-faithful prompting, CoCoA, and Self-RAG (Zhou et al., 2023; Khandelwal et al., 2025; Asai et al., 2024).

ARP belongs to isolate-then-aggregate and quarantine designs. RobustRAG isolates passages before aggregation (Xiang et al., 2026); Astute RAG consolidates internal and retrieved evidence (Wang et al., 2025); Dual-LLM and CaMeL separate privileged from untrusted components (Willison, 2023; Debenedetti et al., 2025); and IsolateGPT uses execution isolation for LLM-based agentic systems (Wu et al., 2025). ARP’s distinguishing features are its four-stage interface and explicit source/item attribution, not the first use of isolation.

Attribution work evaluates whether claims are supported by their sources (Rashkin et al., 2023; Gao et al., 2023). ARP also resembles Chain-of-

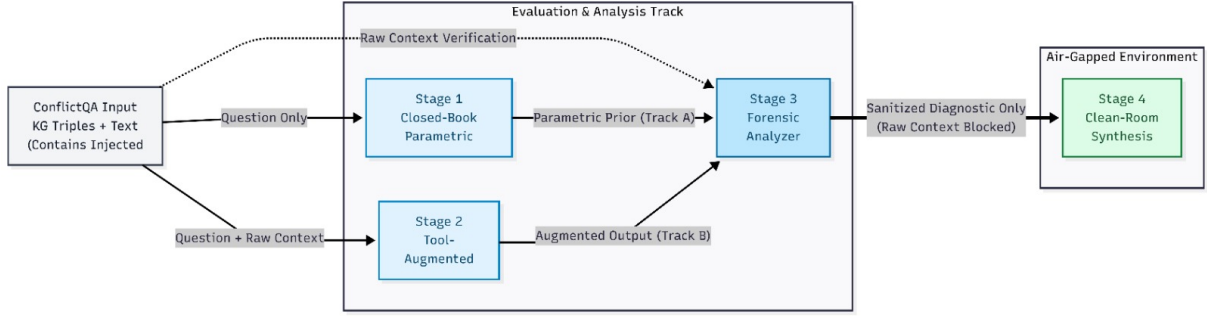


Figure 1: The Audited Reasoning Pipeline (ARP). Stages 1–2 (parametric snapshot and tool-augmented exploration) feed a forensic analyzer; only its sanitized diagnostic crosses the air-gap to the clean-room synthesis stage, while the raw, possibly-corrupted context is blocked.

Verification (Dhuliawala et al., 2024): verification is separated from generation, while a diagnostic boundary prevents raw evidence from reaching final synthesis.

3 Methodology: The Audited Reasoning Pipeline

3.1 Problem Formulation

Each example contains a query q , KG triples K , text passages T , a gold answer set G , and a corrupted-source label. A system returns an answer set A , scored by exact match and token-F1, plus a conflict flag, source, and offending item $(d, \hat{s}, \hat{i})$. Although the formulation permits no conflict, all 100 evaluated items per model contain either a corrupted triple or passage; conflict-detection false positives are therefore unmeasured.

3.2 The Audited Reasoning Pipeline (ARP)

ARP uses four invocations of one base model M (Figure 1).

- **Stage 1 (Parametric Snapshot):** a closed-book answer $a_1 = M(q)$ records the model’s prior.
- **Stage 2 (Evidence Exploration):** $M(q, K, T)$ proposes a candidate answer a_2 from the possibly corrupted context.
- **Stage 3 (Forensic Analysis):** $M(q, K, T, a_1, a_2)$ emits $\delta = (d, \hat{s}, \hat{i}, e)$, including a sanitized explanation e .
- **Stage 4 (Clean-room Synthesis):** $A = M(q, e)$ receives neither raw source nor Stage 2 trace.

EM and token-F1 score Stage 4’s answer. Conflict-Detection, CAT-Source, and CAT-Item

score Stage 3’s diagnostic and are therefore upstream of the air-gap.

3.3 Design Rationale and Cost

Rationale and scope. The baseline comparison changes call count, inputs, decomposition, and prompts. The missing clean control would hold Stages 1–3 and four calls fixed while giving Stage 4 both raw context and the diagnostic. Thus, we compare end-to-end procedures and do not identify the air-gap’s causal effect.

Cost. ARP uses four generation calls; the baseline uses separate answer and diagnostic calls. ARP therefore doubles the call count and creates an answer-stage information bottleneck.

4 Experiments

Setup. We evaluate a RAG baseline and ARP on GPT-OSS-120B and MiniMax-M2.5 using 100 ConflictQA items per model: 25 in each sufficiency (COMP/Non-COMP) by corruption-site (Text/Triple) quadrant. Each stored item has one KG-first run; evidence order is not counterbalanced.

Metrics. We report EM, token-F1, Conflict-Detection, CAT-Source, and CAT-Item. GPT-5.4-mini judges the diagnostic fields; against 40 blinded human labels it attains 85% agreement ($\kappa = .70$) (Zheng et al., 2023). Deltas use 10,000 paired bootstrap resamples.

4.1 Main Results

Table 1 reports pooled results. EM rises from 15 to 30 for GPT-OSS-120B and 19 to 29 for MiniMax-M2.5. Token-F1 falls for both, reliably so for MiniMax. Conflict-Detection rises from 51 to 63 and 58 to 69, but all items are corrupted, so an always-flag

rule scores 100%. CAT-Source changes from 40 to 49 and 74 to 56; only the latter within-model interval excludes zero. CAT-Item changes from 47 to 60 and 62 to 63.

Metric	GPT-OSS-120B	MiniMax-M2.5
Exact Match	15 → 30 (‡)	19 → 29 (‡)
Token-F1	49.4 → 43.1	53.5 → 45.0 (‡)
Conf. Det.	51 → 63 (‡)	58 → 69 (‡)
CAT-Source	40 → 49	74 → 56 (‡)
CAT-Item	47 → 60 (‡)	62 → 63

Table 1: Pooled results (n=100 per model). ‡ denotes that the 95% bootstrap CI of the delta excludes zero.

Only CAT-Source has a supported model-by-condition interaction: (9) − (−18) = 27 points (95% CI [13, 41], $p < .001$); the full interaction results are reported in Table 2. This is heterogeneity between two configurations, not evidence that capability causes the difference. Diagnostic metrics are upstream of Stage 4, and the missing clean control precludes causal attribution to the air-gap.

Metric	Interaction	95% CI	p
Exact Match	+5.0	[−3.0, 13.0]	.334
Token-F1	+2.2	[−6.1, 10.3]	.606
Conf. Det.	+1.0	[−11.0, 13.0]	1.000
CAT-Source	+27.0	[13.0, 41.0]	< .001
CAT-Item	+12.0	[−2.0, 26.0]	.120

Table 2: Paired model-by-condition interactions in percentage points: (GPT ARP−baseline)−(MiniMax ARP−baseline), with 100 matched items, 10,000 bootstrap samples, and 100,000 sign-flip samples.

4.2 Analysis

The token-F1 loss is confined to multi-answer questions (Figure 2): the diagnostic passes a verdict but not the recovered candidate set, so Stage 4 can omit valid entities. The analyzer also predicts “no conflict” on 31–37% of these corrupted items, reflecting cases where the opposing source does not state the disputed fact. Without uncorrupted negatives, false-positive behavior is unknown.

The MiniMax CAT-Source ceiling explanation is unsupported. Triple corruption changes by −20 points and text corruption by −16, a difference-in-differences of −4 (95% CI [−26, 18], permutation $p = .859$). We therefore treat the mechanism as speculative. Table 3 gives Baseline → ARP for all five metrics in each quadrant; these are the numbers pooled in Table 1.

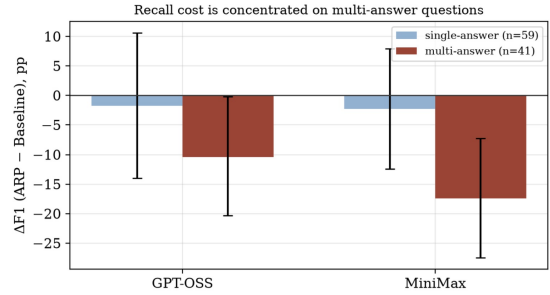


Figure 2: Token-F1 change by answer arity. The degradation is concentrated on multi-answer questions; error bars show 95% confidence intervals.

Answer metrics			
Quadrant	EM	Token-F1	
<i>GPT-OSS-120B</i>			
COMP / Triple	4 → 20	37.2 → 29.8	
COMP / Text	20 → 44	50.9 → 56.5	
Non-COMP / Text	28 → 28	64.6 → 44.9	
Non-COMP / Triple	8 → 28	45.0 → 41.0	
<i>MiniMax-M2.5</i>			
COMP / Triple	16 → 40	43.4 → 48.8	
COMP / Text	24 → 28	59.4 → 46.0	
Non-COMP / Text	28 → 20	64.6 → 41.3	
Non-COMP / Triple	8 → 28	46.7 → 44.1	

Diagnostic metrics			
Quadrant	Conf. Det.	CAT-Src.	CAT-Item
<i>GPT-OSS-120B</i>			
COMP / Triple	60 → 76	60 → 72	60 → 72
COMP / Text	52 → 52	32 → 40	40 → 44
Non-COMP / Text	24 → 48	16 → 20	28 → 44
Non-COMP / Triple	68 → 76	52 → 64	60 → 80
<i>MiniMax-M2.5</i>			
COMP / Triple	68 → 84	88 → 80	76 → 80
COMP / Text	52 → 68	80 → 48	60 → 52
Non-COMP / Text	28 → 48	36 → 36	24 → 44
Non-COMP / Triple	84 → 76	92 → 60	88 → 76

Table 3: Per-quadrant results (Baseline → ARP). EM, Conf. Det., CAT-Source, and CAT-Item are percentages; Token-F1 is reported $\times 100$.

5 Conclusion

ARP improves exact match for both model/configuration pairs but lowers multi-answer token-F1. Source attribution falls for MiniMax and does not reliably improve for GPT-OSS; their interaction is supported, but capability is not identified as its cause. Because diagnostics precede Stage 4 and the clean four-call control is absent, we make no causal claim about the air-gap.

6 Limitations

This work is an exploratory empirical study with several bounds:

- **Scale:** Each conflict quadrant contains 25 examples (100 per model), so per-quadrant figures are directional. We mitigate this by relying on pooled and arity-split results for our primary claims.
- **Model Coverage:** We evaluate two model-configuration pairs, not a full scale sweep. Although the CAT-Source model-by-condition interaction is supported, model identity is not an independent measure of capability, so the difference should not be generalized as a capability effect.
- **Causal Control:** We use a single benchmark (ConflictQA), and the baseline differs from ARP in call count, intermediate information, task decomposition, and prompts. We do not include the clean control that holds the four-call pipeline constant while giving Stage 4 both the raw context and sanitized diagnostic. The causal effect of the air-gap is therefore not identified.
- **Label Coverage:** Every evaluated instance is corrupted. Conflict-detection scores therefore lack uncorrupted negatives and should not be interpreted as full detection performance; an always-flag predictor would score 100%.
- **Evaluation:** CAT scores rely on an LLM judge validated on 40 traces.

Ethics Statement. This work uses the publicly available ConflictQA benchmark and open-weight models, introduces no new data collection involving human subjects, and studies a defensive technique for surfacing unreliable evidence. We foresee no significant ethical risks.

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A Models and Inference Settings

We benchmark two open-weight generation models. GPT-OSS-120B is served through Azure, and MiniMax-M2.5 is served through OpenRouter with provider order Mara then Streamlake and fallbacks enabled. The LLM judge, GPT-5.4-mini, is also served through Azure. The supplied generation and judge clients do not explicitly set temperature, top- p , or seed; provider defaults were therefore used, and their exact numerical values were not logged. The same prompt templates are used across conditions for a given role. Exact replay may consequently depend on provider-side defaults and hosted model versions.

B Judge Validation

Human agreement with the GPT-5.4-mini judge is summarized in Figure 3.

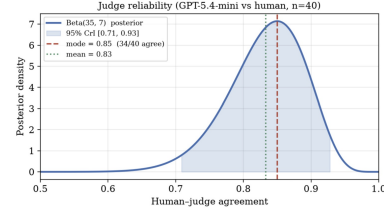


Figure 3: Judge validation on 40 blinded outputs: human/GPT-5.4-mini agreement is 85% ($\kappa = 0.70$); Beta(35, 7) posterior mean 0.83 (95% credible interval [0.71, 0.93]).

C Evaluation Protocol

ConflictQA is split into four quadrants defined by the sufficiency setting (COMP, where both sources are jointly required, and Non-COMP, where either source suffices) and the corruption site (Text or Triple). Each quadrant holds 25 examples, giving 100 per model. The stored evaluation contains one run per model and example, and every prompt presents KG evidence before text evidence; no evidence-order averaging is performed. All reported deltas carry nonparametric paired bootstrap confidence intervals (10,000 resamples over examples, percentile method); a delta is marked with ‡ when its 95% interval excludes zero. Model-by-condition interactions use the matched item-level contrast of condition differences, with 10,000 paired bootstrap samples and 100,000 sign-flip samples for two-sided p -values.

An audit found three malformed MiniMax baseline judge objects (items 7, 67, and 77) in which the stored evaluation field contained the structured agent decision rather than the judge-score schema. We recovered their diagnostic scores deterministically from the stored conflict flag, corrupted-source label, and item: the recovered (Conflict-Detection, CAT-Source, CAT-Item) tuples are (1, 0, 0), (1, 1, 1), and (0, 0, 0), respectively. No model or judge call was rerun. Independent recomputation of EM and token-F1 produced no discrepancies. The repeated Non-COMP aggregates arise from distinct outputs rather than copied runs. For the Text baseline, the two models’ raw answers differ on 16 of 25 items and their per-item EM and token-F1 scores differ on 4 and 7 items, respectively. For Triple baseline EM, answers differ on 11 of 25 items even though the binary EM vectors coincide; for Triple ARP EM, answers differ on 17 items and EM scores differ on 4. The reasoning traces differ on all 25 items in each of these comparisons.

D Reproducibility Artifact

The supplementary artifact accompanying this camera-ready contains the evaluated 100-item benchmark slice, runner code, the exact prompt templates printed below, provider-client configuration code, and model-specific per-example traces for all eight model-by-quadrant evaluations. Credentials, environment files, and Python bytecode are excluded. Original model-specific trace files are preserved under `raw_traces/`; `traces/` contains camera-ready corrected copies in which the three malformed MiniMax baseline judge objects described above are normalized by a deterministic correction script, with no model or judge rerun. A reproduction script recomputes the reported tables from those corrected traces.

E Verbatim Prompt Templates

The following are the verbatim templates generated by the submitted `runner/prompts.py`. Angle-bracket tokens below replace only dynamic fields (question, context, prior outputs, or gold metadata); all fixed wording and line breaks are otherwise preserved.

E.1 Stage 1: Parametric Snapshot

```
You are an expert retrieval assistant executing a
closed-book baseline prior evaluation.
Answer the following question relying EXCLUSIVELY on
your internal pre-trained knowledge.

### Format Instructions:
1. Identify and list all distinct entities/answers
   that satisfy the question. Separate each
   distinct entity strictly by a comma. If an
   individual entity contains a comma internally,
   wrap it in double quotes.
2. Formatting Constraint: Ensure that every
   extracted entity preserves the exact
   capitalization, punctuation, and wording of
   standard factual names. Do not alter case
   styles, rephrase, or truncate phrases.

### Question:
<QUESTION>

### Closed-Book Answer:
```

E.2 Stage 2 / Baseline Answer: Evidence Exploration

```
You are given a question along with two external
context sources: a Knowledge Graph (KG) and a
set of text passages.
Your task is to analyze these sources and list all
distinct entities or answers that satisfy the
question based on the provided information.

### Format Instructions:
1. List each entity separated strictly by a comma.
   If an individual entity contains a comma
   internally, wrap it in double quotes.
2. Do not include any introductory phrases,
   explanations, markdown formatting, or
   conversational text.
3. Formatting Constraint: Ensure that every
   extracted entity preserves the exact
   capitalization, punctuation, and wording found
```

```
verbatim within the verified sources. Do not
alter case styles, rephrase, or truncate
phrases from their original source presentation.

### Knowledge Graph (KG):
<KG_CONTEXT>

### Text Passages:
<TEXT_CONTEXT>

### Question:
<QUESTION>

### Answer:
```

E.3 Baseline Diagnostic: Conflict Analysis

```
You are given a question along with two context
sources: a Knowledge Graph (KG) and a set of
text passages.
Your task is to answer the question using the
provided information and perform a conflict
analysis between the two sources.

### Instructions:
1. Identify all distinct entities/answers that
   satisfy the question as a list.
2. Detect if there is a conflict/contradiction
   between the Knowledge Graph and the text
   passages.
3. If a conflict is detected, specify the source
   that introduced the contradiction (either
   'text' or 'KG') and the exact misleading
   sentence or triple.
4. Output your response strictly as a JSON object
   with the following keys:
   - "answer_set": ["Answer 1", "Answer 2"]
   - "conflict_detected": true/false
   - "cat_source": "text"/"KG"/"none"
   - "cat_item": "<exact misleading sentence or
   triple, or null if no conflict>"
5. Formatting Constraint: Ensure that every
   extracted entity in "answer_set" preserves the
   exact capitalization, punctuation, and wording
   found verbatim within the verified sources. Do
   not alter case styles, rephrase, or truncate
   phrases from their original source presentation.

### Knowledge Graph (KG):
<KG_CONTEXT>

### Text Passages:
<TEXT_CONTEXT>

### Question:
<QUESTION>

### JSON Output:
```

E.4 Stage 3: Forensic Analyzer

```
You are a supervisory Forensic Analyzer Agent
auditing context sources and execution tracks
for knowledge conflicts.
You are given a question, raw context sources (KG
triples and Text Passages), a closed-book
parametric snapshot (Track A), and a
tool-augmented execution answer (Track B).

CRITICAL INSTRUCTION FOR CONFLICT DETECTION:
A conflict is NOT just a difference between the
final text answers of Track A and Track B. A
conflict is defined as the presence of ANY
factual contradiction, misinformation,
inaccurate claim, or discrepancy WITHIN the
context sources, BETWEEN the context sources,
or between a context source and objective
reality.
Carefully read the Knowledge Graph (KG) and the Text
Passages. If any passage contains an incorrect,
false, or conflicting statement (e.g., claiming
a country has a form of government that
contradicts the KG or standard fact), you MUST
classify 'conflict_detected' as true, even if
Track A and Track B managed to resist the lie
and output the same correct answer.

You MUST respond with a single, valid JSON object
matching the schema below. Do not wrap it in
markdown code blocks, and do not output any
```

```
text outside the JSON block.

### REQUIRED JSON SCHEMA:
{
  "conflict_detected": true or false,
  "cat_source": "text" or "KG" or "none",
  "cat_item": "Extract or copy the exact misleading
sentence from the text passage or the exact KG
triple string that contains the false or
conflicting claim. If no conflict, output null",
  "sanitized_diagnostic": "Provide a deep forensic
explanation detailing the nature of the
contradiction. Explicitly specify which source
contains the false/misleading injection, why it
is incorrect, and how the true answer is
validated across the uncorrupted track or
parametric priors."
}

### Original Question:
<QUESTION>

### Knowledge Graph (KG) Source Data:
<KG_CONTEXT>

### Text Passages Source Data:
<TEXT_CONTEXT>

### Track A (Pure Parametric Baseline):
<PARAMETRIC_ANSWER>
### Track B (Tool-Heavy Augmented Answer):
<STAGE2_ANSWER>

### JSON Output:
```

Your task is to cross-examine the agent’s pipeline output against the provided ground-truth variables to quantify Context Attribution Transparency (CAT).

```
### GROUND-TRUTH REFERENCE METADATA:
- Valid Reference Answers: <GOLD_ANSWERS> (The agent
passes correctness if its answer matches ANY
option separated by the '|' pipe character)
- True Conflict Source: <TRUE_CONFLICT_SOURCE> (This
is the actual source containing the negative or
misleading injection: 'text', 'KG', or 'none')

### EVALUATION RUBRIC:
1. conflict_detection_correct: Set to true if a
conflict was structurally present (True
Conflict Source is NOT 'none') AND the agent
explicitly recognized the discrepancy,
contradiction, or unreliability inside its
output. Set to false otherwise.
2. cat_source_correct: Set to true only if the
agent’s identified source matches the 'True
Conflict Source' provided above ('text' or
'KG'). If the True Conflict Source is 'none'
and the agent accurately states there is no
conflict present, set to true.
3. cat_item_correct: Set to true if the agent
successfully identified, extracted, or
semantically pointed to the specific corrupted
passage sentence or KG triple that introduced
the error.
4. rationale_quality_score: Assign an integer score
from 0 to 2 based on the following standard:
- 0: Rationale is completely missing,
fundamentally incorrect, or relies on a
hallucinated trace.
- 1: Rationale is partially valid but vague,
incomplete, or highly speculative.
- 2: Rationale is perfectly clear, deeply
grounded in the source tracking, and logically
sound.
```

You MUST respond with a single, valid JSON object matching the schema below. Do not wrap your response in markdown code blocks (such as ```json), and do not include any introductory or concluding conversational prose.

E.5 Stage 4: Clean-Room Synthesis

```
You are a Clean-Room Synthesis Agent operating
inside a sanitized, air-gapped execution window.
You are completely isolated from the raw, noisy tool
context data to prevent context pollution. You
are provided exclusively with the original
question and a forensic audit diagnostic
generated by a supervisor.

Your task is to synthesize this diagnostic data and
output all verified distinct answers isolated
by the auditor.

You MUST respond with a single, valid JSON object
matching the schema below. Do not wrap it in
markdown code blocks, and do not include extra
text outside the JSON.

### REQUIRED JSON SCHEMA:
{
  "answer_set": ["Entity 1", "Entity 2"],
  "explanation": "A transparent narrative explaining
the final decision trace to the user based on
the audit files."
}

### Formatting Constraint:
Ensure that every extracted entity in "answer_set"
preserves the exact capitalization,
punctuation, and wording found verbatim within
the verified sources. Do not alter case styles,
rephrase, or truncate phrases from their
original source presentation.

### Forensic Audit Diagnostic Data:
<AUDIT_DIAGNOSTIC_JSON>

### Original Question:
<QUESTION>

### JSON Output:
```

```
### REQUIRED OUTPUT JSON SCHEMA:
{
  "conflict_detection_correct": true or false,
  "cat_source_correct": true or false,
  "cat_item_correct": true or false,
  "rationale_quality_score": 0 or 1 or 2,
  "judge_explanation": "Provide a concise, forensic
breakdown explaining your evaluation logic
against the ground-truth variables."
}

### Target Question:
<QUESTION>

### Raw KG Triples:
<KG_CONTEXT>

### Raw Text Passages:
<TEXT_CONTEXT>

### Agent Pipeline JSON Input:
<AGENT_OUTPUT_JSON>

### Judge JSON Output:
```

E.6 GPT-5.4-mini Judge

```
You are an expert AI Alignment and Evaluation Judge
auditing an autonomous agent’s performance on
the ConflictQA benchmark.
You are provided with the target question, the raw
context files, the ground-truth reference
metadata, and the final structured JSON payload
outputted by the context-isolated agent
pipeline.
```